NLP AND BIG DATA · FINAL PROJECT

Torque Diagnostics architecture

A document lake plus a hybrid retrieval answer path. One worker builds the index. The chat request only reads it.

Ingest is batch | Answer is interactive | Low confidence refuses

S3 + Parquet
System of record

BM25 + vectors
How a fault code is found

NLI gate
Whether steps are shown

Ingest path

A manual is written once to S3, then parsed into page-level Parquet. Polars on the worker writes the code map, including the tracing page on each fault row, and the hybrid index. The chat service never parses a PDF.

Upload PDF or photo → **FastAPI** → **S3 raw zone** → **SQS** → **Docker worker** → **Parquet lake**

then **Hybrid index**

Answer path

A typed fault or a photo of an HMI is turned into entities, retrieved from that family's index, and scored before any step is shown. The tracing page comes back on the chunk row.

Text or photo → **NER on the question** → **BM25 + embeddings**

then **Answer from passages** → **NLI + coverage** → **Cite or refuse**

Where each course concept sits

Concept	Component	Tool	Why this one
Data lake, immutable objects	Raw manuals	S3	Original PDF is never edited in place
Columnar storage, partition pruning	Parsed pages and chunks	Parquet, Hive-style asset_family prefixes	A query for ACS880 skips every other drive family
Table build on the worker	Code map and tracing page	Polars	Fault 2310 carries its tracing page on the chunk row
Single-node analytics	Eval and coverage SQL	DuckDB on the Parquet	Which fault codes have no indexed procedure
Event-driven ingest	Start work when a PDF lands	S3 event to SQS	OCR of a 400-page scan exceeds a Lambda timeout

Concept	Component	Tool	Why this one
Reproducible runtime	API and worker	Docker on EC2, IAM role	Same image on a laptop and in the account, no keys on disk
Lexical match	Fault codes, part numbers	BM25	2310 must match the code, not a nearby vector
Dense retrieval	Symptom wording	Hugging Face embeddings	Overcurrent and excess current should meet
Entity extraction	Asset, fault, part, procedure	Token classifier	Filters the index before ranking
Grounded generation	Technician steps	RAG prompt over retrieved rows	No step that is absent from a passage
Entailment	Confidence	MNLI cross-encoder	Contradiction forces a refusal

Confidence gate

Three signals, one label. A contradiction on the entailment check overrides a strong retrieval score.

Label	Rule	What the technician sees
High	Strong hybrid hit, entailment, and every question entity in the evidence	Steps, page citation, highlighted rows
Medium	Partial entity match, or entailment is neutral	Steps with caveats and the extra passages
Low	Weak retrieval, or the NLI model contradicts the draft	Passages only. No procedure.

On the request path

FastAPI, NER, hybrid search, generation, NLI score. Target is a short answer while the machine is down.

On the worker

OCR, Polars, the Parquet write, and the hybrid index. The citation is the explanation the technician sees.

What the request is allowed to do

The chatbot reads the code map, the BM25 index, and the embedding index that the worker already built. DuckDB reads the same Parquet for coverage checks. The request does not rebuild the lake.